

A trial-level recurrent neural network for CenterTask: architecture and design decisions

Paola Castillo

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Abstract

This document specifies a small recurrent neural network (“tiny RNN”, in the sense of Ji-An et al.) that predicts, trial by trial, the category chosen by a rhesus monkey in a length-categorization task. The network receives the previous choice, the previous bar length, the previous outcome, the current bar length and the rule cue, keeps a hidden state of one to four variables, and reads the choice out through two decision criteria on the length axis. It is trained by maximum likelihood of the animal’s choices and compared, at matched number of state variables, with hand-written cognitive models (model-free reinforcement learning, model-based reinforcement learning, a Bayesian observer and a criterion-shift model). Every decision below is tied to what the task exports and to what the first thirteen sessions show. The within-trial networks (task network and motor network) are specified in a companion document (`RNN_DESIGN.md`).

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1 Purpose

One model is asked three questions.

1. **How much does the past weigh?** The categorization rule is fixed and the reward is a deterministic function of the stimulus, so no previous trial carries information about the correct answer to the current one. Any influence of the past is a bias. Does it exist, how large is it, and what shape does it have?
2. **What is the form of the update rule?** Compared with a criterion model written by hand, with the same number of state variables, does the network find a dynamics the hand-written model cannot express?
3. **How does the 2- versus 3-category rule enter?** All at once, through the cue, or gradually, through feedback?

What the network does *not* do: it does not model anything inside a trial (decision time, trajectory). That is the job of the task network and the motor network of the companion document.

2 Why three networks, and what each one reads

The project has three recurrent networks because the behaviour has two time scales and, inside the trial, two functions. Across trials, the question is how the choice on one trial depends on the trials before it. Inside a trial, the question is first how the category is decided over the 1.5 s from bar onset to go, and then how that decision becomes a joystick movement. Each of the three has its own observable to explain: the choice, the choice together with its timing, and the cursor trajectory. Each therefore reads a different file and a different set of columns (Table 1; the same three are drawn unrolled in Figure 8).

This document specifies network 1 only. Networks 2 and 3 are specified in the companion document `RNN_DESIGN.md`; they appear here only where they connect to network 1 (Sections 8 and 14).

Table 1: The three networks: what each one is for, what it reads, and what it predicts. Column names are those of the task’s export files.

	1. Trial-level tiny RNN	2. Task network	3. Motor network
Question	How does the choice on trial t depend on the trials before it, and on the rule cue?	How is the category decided over the 1.5 s of a trial, and when?	How does the decision state become a cursor trajectory?
Specified in	this document	RNN_DESIGN.md, Section 3	RNN_DESIGN.md, Section 4
One step is	one trial	10 ms	10 ms
One sequence is	one session (240–465 trials)	one trial (≈ 335 steps, bar onset -300 ms to go $+1500$ ms)	the same trial, aligned step by step with network 2
File read	trial_data_<runTag>.csv only (one row per trial)	trajectory_<runTag>.csv (one row per cursor sample, with Time_ms and Epoch) joined to trial_data on (Block, TrialNumInBlock, Attempt)	the same join as network 2
Input features	BarSizeVA_deg of rows t and $t - 1$; ChosenTarget, IsCorrect of row $t - 1$; NumCategories of row t ; optionally DecisionTime_s of row $t - 1$	BarSizeVA_deg switched on during Epoch = 4 (BAR); a rule channel from NumCategories switched on from Epoch = 5 (CUE); a go channel from the onset of Epoch = 6	the bottleneck $b(t)$ read from network 2’s hidden state; DirectionChosen as a one-hot switched on at go; the go channel
Target	ChosenTarget of row t ; optionally DecisionTime_s of row t	ChosenTarget of the trial, repeated on every step after go (earlier steps masked); DecisionTime_s as the check on the read-out crossing time	X_px, Y_px at every step, centre-relative and divided by the 406 px centre-to-target distance
Loss	cross-entropy of the choice; optional Gaussian on log decision time	cross-entropy on post-go steps	squared error on position, valid steps only
What is <i>not</i> used	any trajectory file; within-trial timing other than DecisionTime_s	the previous trials (unless network 1’s state is passed as $h(0)$)	the bar length directly (it arrives only through $b(t)$)

Two consequences of this split are worth stating once. First, network 1 never opens a trajectory file: everything it needs is in the trial table, one row per trial. Second, networks 2 and 3 never look across trials: a trial is a complete sequence for them, and the only way history reaches them is through the optional initial state supplied by network 1.

3 The task and its outputs

3.1 Trial structure

Each trial of CenterTask (CenterOutTask.m) shows a horizontal bar of one of twelve lengths (4.0–7.8 degrees of visual angle), then a colour cue, then two or three coloured targets at cardinal positions that are reshuffled on every trial. The subject moves a joystick cursor from the centre to the target whose colour matches the bar’s category and is rewarded if the category is correct. Figure 1 shows the category structure; Figure 2 the on-screen sequence.

3.2 What the task writes

For every session the task writes trial_data_<runTag>.csv (one row per trial), two trajectory exports at the cursor sampling rate (about 9 ms), a per-condition summary and a text report. Only

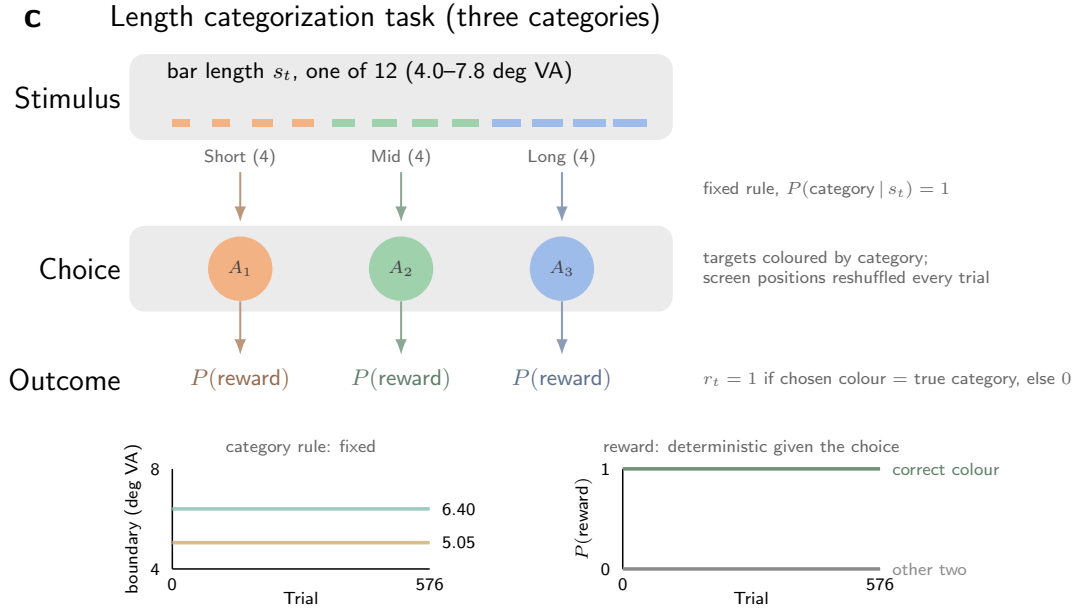


Figure 1: Task structure in the three-category framing. Twelve bar lengths, four per category, map deterministically onto Short, Mid and Long. The category boundaries (5.05 and 6.40 deg VA) are the nominal midpoints between adjacent lengths of different categories; they never change within or across sessions. Reward is deterministic given the choice. In the two-category framing (session mode `2cat`) the split is lengths 1–6 versus 7–12, with a single boundary at 5.85, and only the Short and Long targets are shown.

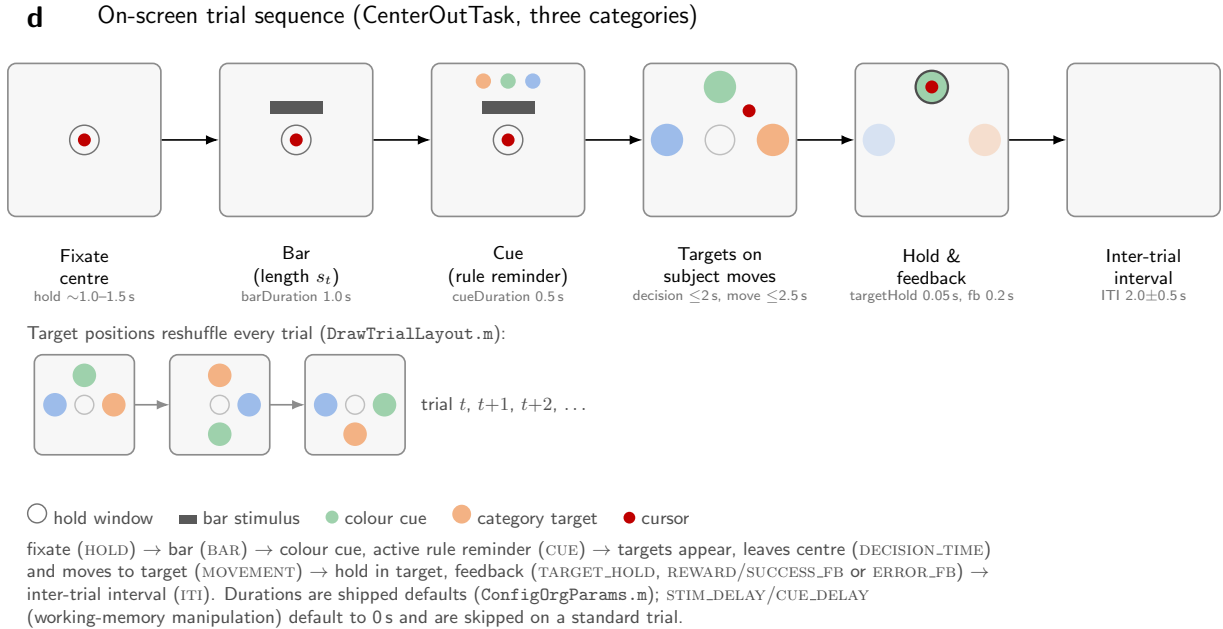


Figure 2: On-screen sequence of one trial, following the epoch order in `TaskEpoch.m`: fixation at the centre, bar (1.0 s), colour cue (0.5 s, two or three dots that remind the subject of the active rule), target onset (go), movement, hold and feedback, inter-trial interval. The bar-to-go interval is essentially fixed (1.55 s, range 35 ms). Target positions reshuffle every trial, so repeating a category and repeating a direction are dissociable.

the trial table is used here. Its relevant columns are listed in Table 2.

Table 2: Columns of `trial_data_*.csv` used by the model and their encoding.

Model variable	Column	Encoding
s_t	BarSizeVA_deg	scalar, normalised to $[-1, 1]$ over 4.0–7.8
a_t	ChosenTarget (1 Short, 2 Mid, 3 Long, 0 none)	ordinal label 0/1/2; none $\rightarrow$ masked
r_t	IsCorrect	binary
cue_t	NumCategories (2 or 3; absent in old schema $\rightarrow$ 3)	one-hot, 2 channels
order	Block, TrialNumInBlock	defines the sequence within a session
retry	Attempt	only Attempt = 1 is kept

The history inputs are the same variables shifted by one trial: a_{t-1} , s_{t-1} , r_{t-1} . The first trial of a session receives a “no history” vector (zeros plus an indicator channel), never the last trial of the previous session.

3.3 What the data show

Thirteen sessions (20 August to 5 September 2026, subject ROM) hold 3 866 trials, of which 3 818 have a response. Overall accuracy is 0.765 and is stable across sessions (0.73–0.81). Errors are almost all wrong-target choices (860) rather than early exits (48). Retries are rare (51).

Psychometric function. Table 3 pools the three-category sessions. The subject’s Short/Mid boundary sits below the nominal 5.05: at 4.70 it already answers Mid half of the time. The Mid/Long boundary falls near the nominal 6.40. A network fitted to behaviour must reproduce this asymmetry; a network trained on the correct labels will not.

Table 3: Probability of each response as a function of bar length, three-category sessions pooled ($n \approx 310$ per length).

Length (deg VA)	$P(\text{Short})$	$P(\text{Mid})$	$P(\text{Long})$
4.00	0.98	0.01	0.01
4.10	0.96	0.02	0.02
4.70	0.42	0.52	0.07
5.00	0.23	0.66	0.11
5.10	0.15	0.75	0.10
5.70	0.04	0.79	0.17
6.00	0.03	0.78	0.19
6.10	0.02	0.68	0.30
6.70	0.02	0.19	0.79
7.00	0.00	0.09	0.90
7.60	0.01	0.02	0.97
7.80	0.02	0.03	0.96

Sequential effects are weak. Accuracy is 0.788 after a correct trial and 0.692 after an error ($n = 2953$ and 913). The probability of repeating the previous category is 0.322 after a correct trial and 0.293 after an error, against a chance level of 1/3. The past therefore contributes little beyond a post-error effect. This bounds what the network can find and motivates the small d below.

The alternate regime is the base regime. From 5 September onward sessions run in session mode `alternate`: full 48-trial blocks of two and three categories follow each other (the first such session ran 2, 2, 3, 3, 2, 2, 3). Accuracy was 0.795 in the two-category blocks and 0.744 in the three-category blocks of that session. This is the regime the network is designed for: identical stimuli, a rule that changes within the day, signalled only by the cue, and a block switch every one

to three blocks. The design below assumes a corpus of **alternate** sessions of the same size as the three-category corpus already recorded (12 to 15 sessions, 50 to 70 block switches; the budget is derived in Section 5). The three-category-only sessions of August and early September are kept as auxiliary training data (Section 9.1).

Filters. Table 4 lists what is removed and why. After filtering there are about 3 750 labelled trials in the 12 sessions recorded so far, 190 of them in two-category blocks; the **alternate** corpus grows with every new session.

Table 4: Exclusions.

Filter	Reason	Trials
Attempt > 1	retries re-show a stimulus the subject just failed and distort the psychometric function	51
ChosenTarget = 0	no choice to predict; the trial stays in the sequence as input to the next one but leaves the loss	48
Session 28-Aug	21 trials, accuracy 0.33	21

4 Problem formulation

Each session is one sequence. At step t the network receives $\mathbf{x}_t$, updates its state $\mathbf{h}_t$ and outputs $\hat{\mathbf{y}}_t = P(a_t \mid \mathbf{x}_{1:t})$. This is a many-to-many problem with $T_x = T_y$: one output per input step, with T equal to the number of trials in the session (240–465). It is neither many-to-one nor an encoder–decoder (Figure 3).

The objective is the likelihood of the observed choices,

$$\mathcal{L}(\theta) = -\frac{1}{N_{\text{lab}}} \sum_{\text{sessions}} \sum_{t=1}^T m_t \log P_{\theta}(a_t = a_t^{\text{obs}} \mid \mathbf{x}_{1:t}), \quad N_{\text{lab}} = \sum m_t, \quad (1)$$

where $m_t = 1$ if the trial has a response and **Attempt** = 1, and $m_t = 0$ otherwise. The network is not trained to solve the task; it is trained to predict what the subject did, errors included.

5 Type of network, size, and data budget

This section answers, in plain terms, what kind of recurrent network is used and why, what its size parameter d means, which activation functions it contains, and how many sessions are needed. The formal equations follow in Section 6.

5.1 Which kind of recurrent network

A recurrent network is any model in which a state vector is updated at every step from its previous value and the current input. Three standard cells differ in how the update is written.

- **Vanilla (Elman) RNN.** One equation, $\mathbf{h}_t = \tanh(\mathbf{W}\mathbf{x}_t + \mathbf{U}\mathbf{h}_{t-1} + \mathbf{b})$. Keeping a value and replacing it go through the same weights, so a unit cannot do both well, and gradients through hundreds of steps vanish or explode.
- **GRU (gated recurrent unit).** The same state, plus two gates that decide, per unit and per step, how much of the old value to keep and how much to overwrite (Eqs. (2)–(5)). One state vector, three weight sets.
- **LSTM.** Two state vectors (cell and hidden) and three gates. More parameters, and the “number of state variables” is ambiguous because each unit carries two numbers.

The network is a GRU, not a vanilla RNN. Three reasons. It is the cell used by Ji-An et al., so the comparison with their results is direct. With $d = 1$ or 2 the gates are what let a single unit hold a criterion across trials and still reset it after an error or a cue; a vanilla unit has to choose

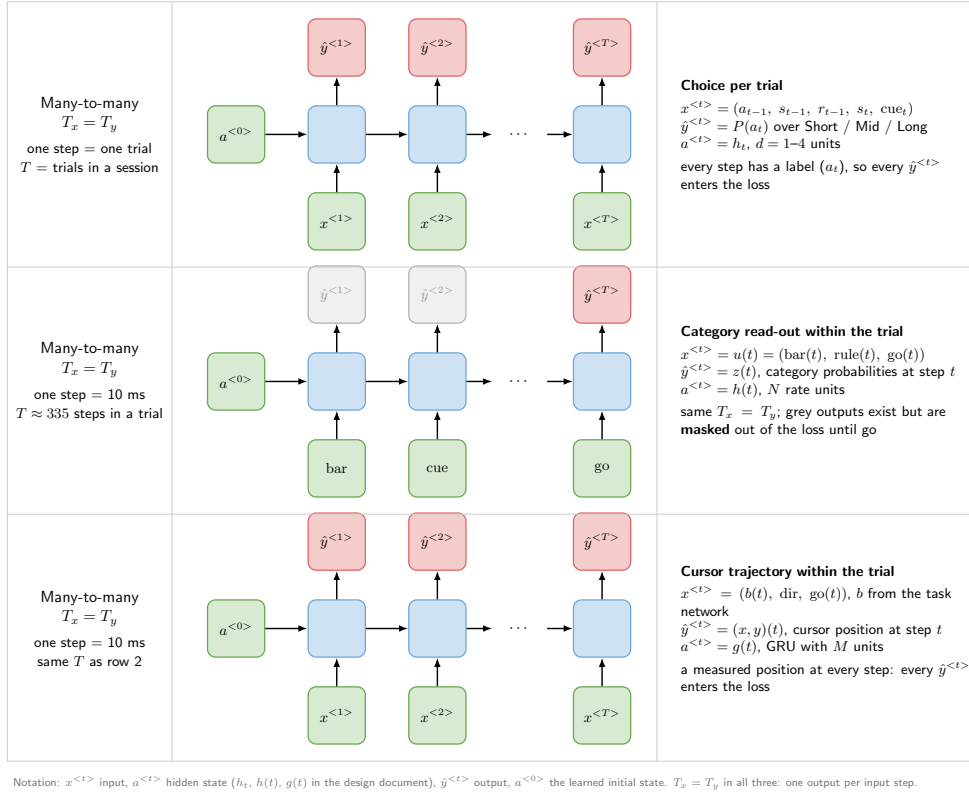


Figure 3: The three networks of the project in the standard many-to-many layout, $T_x = T_y$. Green boxes are inputs $x^{(t)}$, blue boxes the hidden state $a^{(t)}$ (our h_t), red boxes the outputs $\hat{y}^{(t)}$, and $a^{(0)}$ the learned initial state. Top row: this document’s network, one step per trial. Middle and bottom rows: the within-trial task and motor networks of the companion document, one step per 10 ms; grey outputs exist but are masked out of the loss until the go signal.

one behaviour. And it has one state vector, so d is unambiguous. The LSTM is not used because it doubles the state for no gain at this size. A vanilla RNN with the same d is fitted as a control (Section 10): with short memory the gates may not be needed, and if the two cells tie, the vanilla one is preferred because its dynamics are easier to read.

5.2 What d is

d is the number of hidden units, which for a GRU equals the dimension of the state vector $\mathbf{h}_t \in \mathbb{R}^d$, which in turn equals the number of dynamical variables: the number of quantities the model carries from one trial to the next. In the framing of Ji-An et al. this number, not the parameter count, is the measure of model complexity, because it is what a cognitive model also has: the criterion-shift model M2 has $d = 2$ (two boundaries), the Bayesian observer $d = 4$ (mean and width of each boundary), a model-free learner $d = 36$ (one value per length and category). Two models with the same d can be compared on equal terms even if one has twenty parameters and the other eighty, because they remember the same amount and differ only in the form of the update.

With $d = 1$ the entire memory of the animal, as the model sees it, is a single number that moves up and down across trials. That is deliberately restrictive: it is the case in which the whole dynamics can be drawn on a page. The prediction for this task is that $d = 1$ or 2 is enough, because the rule is fixed and sequential effects are weak. $d = 3$ and 4 are fitted to confirm that they do not improve held-out likelihood.

5.3 What a hidden unit is here

A hidden unit is one coordinate of $\mathbf{h}_t$: a scalar that is read from the previous trial, updated with the current inputs, and passed on. It is not a neuron and it is not a feature detector. In this network a unit is expected to behave as a slowly varying bias, for instance the current displacement of a criterion from its resting value. Units have no assigned meaning before training; their meaning is established afterwards by reading how they move the criteria (Section 11). That is the difference between the h_t of the network and the V_t or c_t of a cognitive model, which are named before any data are seen.

5.4 Activation functions, and why those

There are three places where a nonlinearity sits, and only one of them is a choice.

- **Gates: logistic σ .** A gate is a proportion, how much to keep and how much to replace, so it must lie in $(0, 1)$. The logistic function is the standard map onto that interval. Fixed by the GRU definition.
- **Candidate state: tanh.** The state must be bounded, otherwise it can grow without limit over the 300 steps of a session, and it must be symmetric around zero so that a criterion can move in either direction. tanh gives both; a ReLU would give neither. Fixed by the GRU definition.
- **Read-out: logistic or cumulative Gaussian.** Each criterion turns a distance on the length axis into a probability, so the read-out is a sigmoid curve; it is the psychometric function of the model. Logistic and cumulative Gaussian are both used in psychophysics and differ only by a scale factor of about 1.7 in the slope. The logistic is the default; the Gaussian is used when an exact match with the notebook’s psychometric fits is wanted. This is the one choice.

No softmax is used: the ordinal read-out already yields three probabilities that sum to one, and a softmax would discard the ordering of the categories.

5.5 Size of the network and of the data

The network has d units, $k = 9$ (or 10) input channels and between 42 and 189 learned parameters (Table 9). Each session is one sequence of 240 to 465 steps, and gradients are propagated through the whole session. This is small: a fit takes seconds, and the whole grid of models, folds and seeds fits in an afternoon.

The data requirement is set by three needs, and the largest of them decides.

1. **Trials per parameter.** A rule of thumb for a well-conditioned maximum-likelihood fit is at least 20 labelled trials per parameter. With $d = 2$ and 85 parameters that is 1 700 trials, about six sessions. The three-category sessions already recorded cover this for the choice model alone.
2. **Outer folds.** The paired comparison between models uses one difference per test session. Ten differences are the minimum for a bootstrap interval that means anything; twelve to fifteen are comfortable.
3. **Block switches.** The rule questions are answered on the trials after each switch. With segment lengths drawn from one to three blocks and about 320 trials per session, a session holds $\approx 320/\bar{L} - 1$ switches, where $\bar{L}$ is the mean segment length in trials. With segments mostly of one block ($\bar{L} \approx 65$) that is four to five switches per session; with segments averaging two blocks it is two to three. To estimate the post-switch curve on five trials, in both directions (2→3 and 3→2), with a resolution of about one trial in the time constant, at least 50 to 60 switches are needed.

Need 3 dominates. **The target is 12 to 15 alternate sessions, with segment lengths drawn from $\{1, 2, 3\}$ blocks with weights about 0.6/0.3/0.1**, which yields four to five switches per session and 50 to 70 switches in total, and at the same time satisfies needs 1 and 2. Ten sessions is the floor below which the latent-rule version (Section 7) cannot be fitted with confidence; the cue-driven version and the choice-only questions are already answerable with the sessions recorded. The three-category-only sessions add to need 1 but not to needs 2 and 3.

6 Architecture

6.1 Overview

Figure 4 places the network next to a cognitive model in the framing of Ji-An et al.: both are dynamical systems with d state variables, the same inputs and the same output. They differ only in who writes the update rule. In the cognitive model the variables have a meaning fixed in advance (here, decision criteria c_t) and the update rule is written by the modeller; in the network the variables h_t have no assigned meaning and the update rule is learned.

6.2 Inputs

The input vector has $k = 9$ channels (Table 5).

Table 5: Input channels $\mathbf{x}_t$.

Channel	Dim.	Content
s_t	1	current bar length, normalised
s_{t-1}	1	previous bar length, normalised
a_{t-1}	3	one-hot of the previously chosen category
r_{t-1}	1	1 if the previous choice was rewarded
cue_t	2	one-hot: two categories / three categories
first	1	1 on the first trial of a session
total k	9	
$\log \text{DT}_{t-1}$ (optional)	1	previous decision time, standardised; Section 8

Design decision: s_t enters twice. The current length is fed both to the recurrent cell and directly to the read-out (Section 6.4). With $d = 1$ or 2, if s_t entered only through the cell, the unit would be spent representing the stimulus and no memory would remain. The direct connection leaves the cell free to represent only what changes from trial to trial.

a

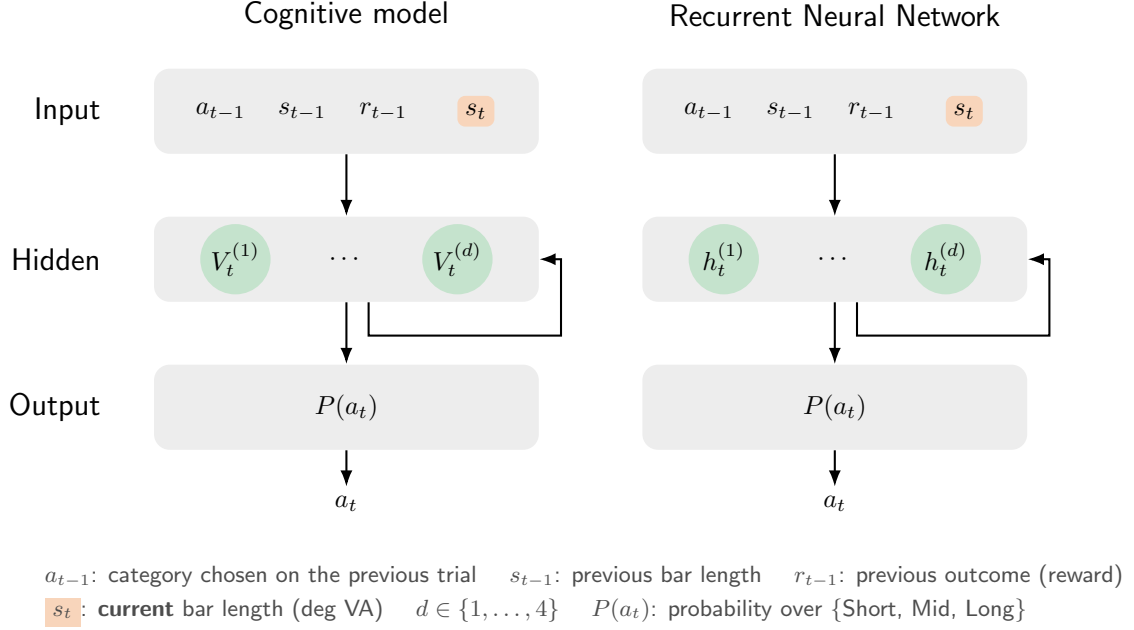


Figure 4: Trial-level model. Left: a cognitive model whose d state variables $V_t^{(i)}$ carry a meaning assigned by the modeller (for this task, decision criteria). Right: the recurrent network, whose d hidden units $h_t^{(i)}$ are interpreted after training. Both receive the previous choice a_{t-1} , the previous bar length s_{t-1} , the previous outcome r_{t-1} and, highlighted, the current bar length s_t , the input the reward-learning tasks of Ji-An et al. do not have. Both output a probability over Short, Mid and Long.

What is left out. `PrevTrialDirection` is logged but not used: without an export of the target layout it cannot be related to the chosen category. It becomes usable once the layout is exported (Section 13).

6.3 State: a gated recurrent unit

The state is updated by a GRU cell (Figure 5):

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1} + \mathbf{b}_r) \quad \text{reset gate} \quad (2)$$

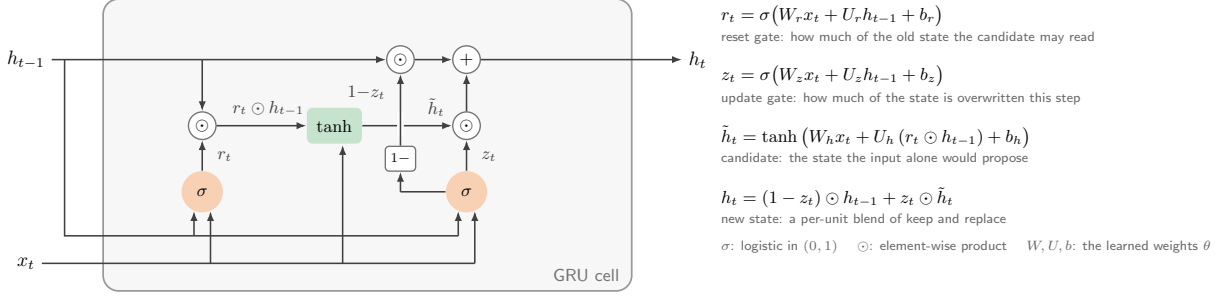
$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1} + \mathbf{b}_z) \quad \text{update gate} \quad (3)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{U}_h (\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \quad \text{candidate} \quad (4)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \quad \text{new state} \quad (5)$$

with σ the logistic function, $\odot$ the element-wise product, and $\mathbf{h}_t \in \mathbb{R}^d$.

One GRU cell: two gates decide how much of the past survives



In the trial-level tiny RNN: $x_t = (a_{t-1}, s_{t-1}, r_{t-1}, s_t, \text{cue}_t)$ and h_t has $d = 1-4$ units. With $d = 1$ the cell has one reset value, one update value and one state, so the whole dynamics can be plotted.

What the gates mean here: $z_t \rightarrow 0$ keeps the criterion where it was (long memory, the Ji-An reward-learning regime); $z_t \rightarrow 1$ rewrites it from the current trial (short memory, the regime the weak sequential effects predict). $r_t \rightarrow 0$ lets the candidate ignore the past, which is how a rule switch can reset the state in one trial.

In the motor network: same cell with $x(t) = (b(t), \text{dir}, \text{go})$ every 10 ms and $M \approx 32$ units; the gates let it hold the pre-go position and then release the movement.

Why a GRU and not a plain RNN: a plain unit is $h_t = \tanh(Wx_t + Uh_{t-1} + b)$, so keeping a value and replacing it both go through the same weights; with one or two units it cannot do both well. The gates give each unit an explicit keep/replace switch that depends on the input, which is what lets a single unit hold a criterion across trials and still reset it after an error or a cue.

Cost: three weight sets instead of one. With $d = 2$ and five inputs that is about 50 parameters, still far below the 3800 trials available.

Figure 5: One GRU cell. The reset gate r_t decides how much of the old state the candidate may read; the update gate z_t decides how much of the state is overwritten; the new state is a per-unit blend of keep and replace. In this task, $z_t \rightarrow 0$ keeps a criterion where it was (long memory, the reward-learning regime of Ji-An et al.), $z_t \rightarrow 1$ rewrites it from the current trial (short memory, the regime the weak sequential effects predict), and $r_t \rightarrow 0$ lets the candidate ignore the past, which is how a rule switch can reset the state in a single trial.

Table 6: Decisions about the recurrent cell.

Element	Decision	Justification
Cell type	GRU	Same as Ji-An et al., so the comparison is clean; with small d the gates let a single unit both hold and reset
Comparison	plain RNN $h_t = \tanh(Wx_t + Uh_{t-1} + b)$, same d	With short memory the advantage of gating may vanish; if the two tie, the plain cell is preferred for interpretability
d	1, 2, 3, 4, one at a time	The main complexity axis. Prediction: 1 or 2 suffice
h_0	learned vector shared across sessions	The start of a session is the same “no history” state everywhere
Activations	σ in the gates, $\tanh$ in the candidate	Fixed by the GRU definition: proportions in $(0, 1)$, a bounded and symmetric state

6.4 Read-out: ordinal with two criteria

The categories are ordered along the length axis, so the output is parameterised as a perceived length and two criteria rather than three free logits:

$$\hat{x}_t = g(s_t, h_t) = s_t + w_g^\top h_t + b_g \quad \text{perceived length} \quad (6)$$

$$c_1(h_t) = c_1^0 + w_1^\top h_t \quad \text{first criterion} \quad (7)$$

$$c_2(h_t) = c_1(h_t) + \text{softplus}(\delta^0 + w_2^\top h_t) \quad \text{second criterion, } c_2 \geq c_1 \quad (8)$$

$$P(a_t \geq k) = (1 - \lambda) F(\beta(\hat{x}_t - c_k)) + \frac{\lambda}{2}, \quad k = 1, 2 \quad (9)$$

and the category probabilities follow from the cumulative ones:

$$P(a_t = \text{Short}) = 1 - P(a_t \geq 1), \quad P(a_t = \text{Mid}) = P(a_t \geq 1) - P(a_t \geq 2), \quad P(a_t = \text{Long}) = P(a_t \geq 2). \quad (10)$$

Table 7: Decisions about the read-out.

Element	Decision	Justification
Form	cumulative ordinal	Same structure as the criterion cognitive model and as the notebook’s psychometric fit, which uses the ordinal splits $P(\text{chosen} \geq k)$
F	logistic by default; cumulative Gaussian as an option	Logit and probit differ by a factor of about 1.7 in slope; the Gaussian matches the notebook exactly
β	learned slope, shared by both criteria	Perceptual sensitivity
λ	learned lapse, bounded in $[0, 0.1]$	Stimulus-independent errors; without it the slope absorbs the lapses
Order $c_2 \geq c_1$	softplus parameterisation, Eq. (8)	Guarantees a non-negative probability for Mid
Two-category blocks	same read-out, nothing masked	$P(\text{Mid}) = P(a \geq 1) - P(a \geq 2)$ vanishes when $c_1 \rightarrow c_2$: the network learns to collapse the criteria onto 5.85

Separation of biases. Whatever enters through g shifts *perception* and affects both rules equally (serial dependence on s_{t-1}). Whatever enters through c_k shifts the *decision* (post-error shift, rule). This separation is written into the architecture and is read directly from the weights $\mathbf{w}_g$ versus $\mathbf{w}_1, \mathbf{w}_2$.

6.5 Activation functions

No softmax is used: the ordinal read-out already produces a distribution that sums to one.

Table 8: Where each activation function sits and whether it is a choice.

Place	Function	Range	Chosen or fixed
Gates $\mathbf{r}_t, \mathbf{z}_t$	logistic	$(0, 1)$	fixed (GRU)
Candidate $\tilde{\mathbf{h}}_t$	tanh	$(-1, 1)$	fixed (GRU)
Perceived length $\hat{x}_t$	linear	$\mathbb{R}$	fixed (a position on the axis)
Criteria c_k	linear	$\mathbb{R}$	fixed
Cumulative probability	logistic or cumulative Gaussian	$(0, 1)$	chosen

6.6 Learned parameters

With $k = 9$ input channels the parameter counts are those of Table 9. With about 3 750 labelled trials there are between 20 and 90 trials per parameter, enough for $d \leq 2$. The $d = 4$ configuration needs stronger regularisation and is included to confirm that it does not improve.

Table 9: Number of learned parameters.

Block	Formula	$d = 1$	$d = 2$	$d = 4$
GRU	$3(dk + d^2 + d)$	33	72	168
$\mathbf{h}_0$	d	1	2	4
Read-out g	$d + 1$	2	3	5
Criteria	$2(d + 1)$	4	6	10
β, λ	2	2	2	2
Total		42	85	189

7 Two ways of knowing the rule

The subject sees two or three dots in the cue and two or three targets on screen. Giving that information to the network is not cheating: it models a subject who reads the cue. Withholding it forces the network to infer the rule from feedback, which is what the networks of Ji-An et al. do in reversal tasks. Both versions are fitted (Table 10, Figure 6).

Table 10: The two rule versions.

Version	cue_t	Models	Distinctive prediction
Cue-driven (main)	input channel	a subject who reads the cue	criteria move on the trial of the block switch
Feedback-inferred	channel off	a subject who infers the rule from errors	criteria drift over several trials until they collapse onto 5.85

The cue-driven version is the ceiling: the latent version cannot beat it, and the question is how close it gets and with what delay. A third reading comes for free: if, even with the cue given, the learned dynamics show gradual carry-over, the subject does not fully use the cue.

Trial-level RNN: how the network knows the rule (2 vs 3 categories)

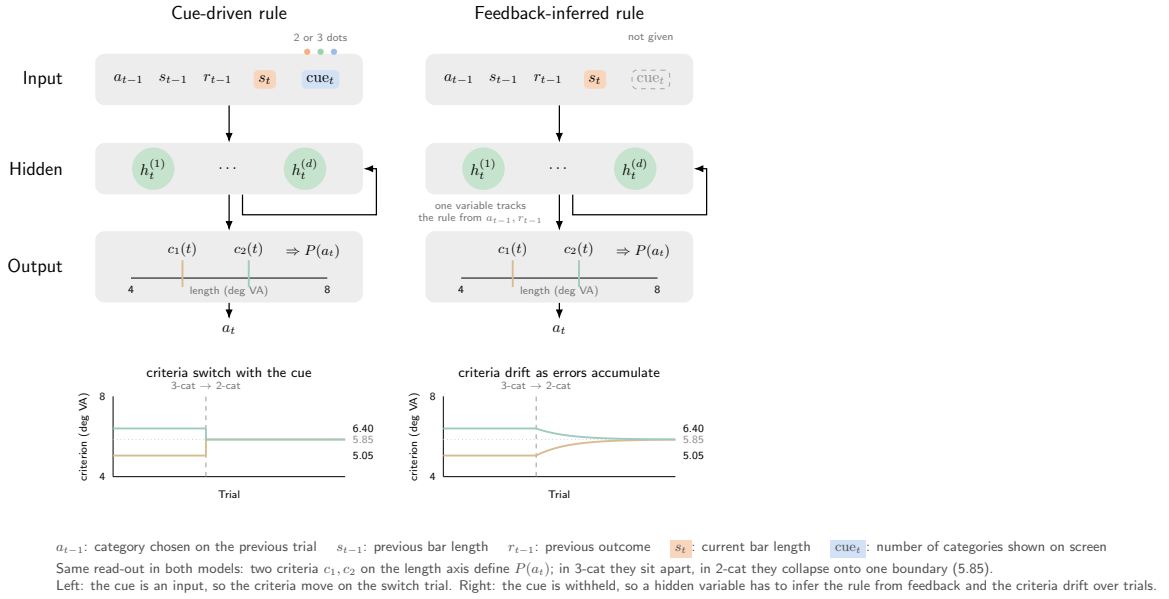


Figure 6: Cue-driven versus feedback-inferred rule. Both networks share the ordinal read-out with two criteria c_1, c_2 on the length axis. Left: the cue is an input, so the criteria can jump from the three-category positions (5.05, 6.40) to the two-category position (5.85) on the switch trial. Right: the cue is withheld, so a hidden variable must infer the rule from a_{t-1}, r_{t-1} and the criteria drift as errors accumulate. The lower plots are the predicted criterion trajectories around a block switch; once fitted, the same plot is produced from the data.

8 Decision time as a second observable

The trial table logs, for every trial with a response, the time from target onset to leaving the centre (**DecisionTime.s**, median 351 ms, 5th–95th percentile 292–492 ms) and the time from leaving the centre to reaching the target (**ExecutionTime.s**, median 183 ms). The first is the time of the decision; the second is motor and is left to the motor network of the companion document. The decision time can enter the model in two ways, which answer different questions.

8.1 The previous decision time as an input

DT_{t-1} is observed before trial t , so it can be an input channel without any leakage. It carries something a_{t-1} and r_{t-1} do not: how sure the subject was. A slow correct choice and a fast correct choice are different histories, and the criterion shift may depend on that. The channel is $\log \text{DT}_{t-1}$, standardised within the training set, set to zero on trials without a response; the input dimension becomes $k = 10$. It is tested with the same ablation as the other channels.

What cannot be done is to use DT_t , the current trial’s time, as an input for predicting a_t : it is part of the same response that is being predicted.

8.2 The current decision time as a second output

The more informative use is to predict choice and decision time jointly from the same state. The decision time is modelled as log-normal with a mean that depends on the distance from the perceived length to the nearest criterion:

$$D_t = - \min_{k \in \{1,2\}} |\hat{x}_t - c_k(\mathbf{h}_t)| \quad \text{proximity to the nearest criterion} \quad (11)$$

$$\mu_t = \mu_0 + \kappa D_t + \mathbf{w}_{\text{DT}}^\top \mathbf{h}_t \quad \text{predicted log decision time} \quad (12)$$

$$\log \text{DT}_t \sim \mathcal{N}(\mu_t, \sigma_{\text{DT}}^2) \quad (13)$$

with learned μ_0 , κ , $\mathbf{w}_{\text{DT}}$ and σ_{DT} . The loss of Eq. (1) gains a second term,

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{choice}}(\theta) + \lambda_{\text{RT}} \mathcal{L}_{\text{DT}}(\theta), \quad \mathcal{L}_{\text{DT}} = \frac{1}{N_{\text{lab}}} \sum m_t \left[\frac{(\log \text{DT}_t - \mu_t)^2}{2\sigma_{\text{DT}}^2} + \log \sigma_{\text{DT}} \right], \quad (14)$$

with λ_{RT} chosen on the validation session so that neither term dominates (start at $\lambda_{\text{RT}} = 1$ with both terms in nats per trial). The same mask m_t applies: trials without a response have no decision time.

Three reasons to do this.

1. **A second observation per trial.** With one categorical label per trial, the choice constrains the hidden state weakly. The decision time is continuous and constrains it far more. The joint fit is expected to identify d and the criterion dynamics with less data.
2. **A direct test of the criterion interpretation.** If c_1 and c_2 are the subject’s boundaries, decision times must peak where $\hat{x}_t$ falls near them. That is the chronometric function of the notebook (Section 1.6), computed with the model’s boundaries instead of the nominal ones. A model that predicts the choice well and the time badly has criteria that are not the subject’s. Equation (11) makes the test explicit: κ should be positive.
3. **A link to the task network.** The within-trial task network produces a decision time as the moment its read-out crosses a threshold. The trial-level μ_t and the within-trial crossing time must agree trial by trial; if they do not, one of the two networks carries a bias the other does not.

To keep the comparison with the cognitive models fair, $\mathcal{L}_{\text{choice}}$ is always reported on its own, and the criterion model M2 receives the same decision-time read-out, Eqs. (11)–(13), which it can because it also has criteria. Models without criteria (M3) get $\mu_t = \mu_0 + \mathbf{w}^\top \mathbf{Q}_t(s_t, \cdot)$.

8.3 Elapsed time as an input

The interval between trials is not constant: 2.0 s after a correct trial, 3.0 s after an error, plus pauses. If the memory of the criterion decays with real time rather than with trial count, a channel with Δt since the previous trial lets the GRU learn a time-dependent forgetting. The value comes from `Time_ms` in the trajectory export, not from the trial table. It is added last, and only if the ablation of r_{t-1} suggests decay.

Order of adoption. First DT_{t-1} as an input, because it is one more ablation and nothing else changes. Then the joint choice–time model, because that is where the scientific gain is. Then Δt , if needed.

9 Training

Table 11: Training protocol.

Element	Value	Justification
Loss	negative log-likelihood per trial, Eq. (1)	It is what is compared across models
Optimiser	Adam, learning rate 10^{-3} , gradient clipping at 1.0	Standard; sequences are long and clipping prevents blow-ups
Batch	one whole session per sequence; batches of 4 sessions	The state must run through the session uncut, so that carry-over across blocks is observable
Early stopping	on validation NLL, patience 50 epochs	Replaces a fixed epoch count; prevents memorisation
Regularisation	L2 10^{-4} on all weights; L1 10^{-4} on $\mathbf{w}_g, \mathbf{w}_1, \mathbf{w}_2$	Pushes state-to-read-out connections that do not help to zero, which simplifies interpretation
Seeds	10 per configuration; best on validation reported, with the spread	With $d = 1$ the loss landscape has local optima
Distillation (optional)	teacher GRU $d = 16 \rightarrow$ student $d = 1, 2$ on the same sequences	Used by Ji-An et al. with scarce data; data are scarce here
Split	nested cross-validation by session, Section 9.1	The unit of independence is the day, not the trial
Report	mean test NLL per trial averaged over folds, bootstrap interval over sessions	

Trials without a response and retries remain in the sequence as input to the following trial; the mask $m_t = 0$ only removes them from the loss. Removing them from the sequence would break the history the subject actually experienced.

9.1 Data split: nested cross-validation by session

Why the unit is the session. The state $\mathbf{h}_t$ runs through the whole session, so consecutive trials are dependent by construction. A day also has effects of its own that are not in the inputs: criterion drift, fatigue, joystick calibration. A split by trial would let the model learn the state of the day from the training trials and use it on the test trials of the same day, which inflates the likelihood and says nothing about generalisation. A split by trial is defensible with tens of thousands of trials per animal, as in the data of Ji-An et al.; it is not with about 300 trials per day.

Corpus. Two sets of sessions enter the fit. The **alternate** sessions are the target corpus: every question of Section 1 is answered on them. The three-category-only sessions are auxiliary: they enter training with $\text{cue}_t = (0, 1)$ throughout, add about 3500 trials of psychometric structure, and are never used as test for any question that involves the rule.

Outer loop: leave one alternate session out. With S **alternate** sessions there are S folds. In each fold one **alternate** session is the test set and is not touched during fitting. The remaining $S - 1$ **alternate** sessions plus all auxiliary sessions form the fitting set. Test NLL per trial, averaged over folds, is the number reported for every model. This measures generalisation to a new day.

Inner loop: one validation session. Inside the fitting set, one **alternate** session is set aside as validation. It drives early stopping and the choice of regularisation strength and seed. It is rotated over the $S - 1$ candidates and the resulting models are averaged in NLL, or, if compute is short, fixed to the session closest in date to the test session. The number of state variables d is *not* selected in the inner loop: d is the quantity of interest, so every value is fitted separately and its outer test NLL is reported.

Paired comparison. All models of Section 10 are evaluated on the same folds. The difference in NLL between two models on each test session gives S paired values; the confidence interval is a bootstrap over those S differences. Day-to-day variation in absolute NLL cancels in the pairing.

Chronological check. In addition to the random leave-one-out, a forward-chaining pass: fit on all sessions before day j , test on day j , for every j from the third **alternate** session onward. If the chronological NLL is worse than the leave-one-out NLL, the subject is still changing between days and the report says so.

Secondary measure: blocks held out within a session. For the dynamics right after a block switch (Section 11, items 1 and 2) the outer folds are complemented by a within-session evaluation: in each **alternate** session, every second two-category block and every second three-category block is masked out of the loss while the network still reads it as input, and the NLL on the five trials after each masked switch is reported separately. This is the masked trial-level split, and it is legitimate here because the question is “within this day, does the model predict the trials that follow a switch?”. It is reported apart from the outer result, never mixed with it.

The day-specific component. The gap between the within-session NLL (blocks held out) and the between-session NLL (day held out) is a direct measure of what is specific to the day and not carried by the inputs. If it is small, the model transfers across days. If it is large, there is a state of the day the network cannot infer from the history, which motivates a per-session parameter or more sessions.

Cost. With $S = 10$ outer folds, four values of d , ten seeds and eight models the grid is about 3 200 fits. Each fit takes seconds on networks of this size.

10 Comparison models

All comparison models receive the same inputs, produce the same output and are fitted with the same loss and the same split. They are compared by held-out NLL, at matched d where applicable. Figure 7 shows how each one operationalises the same problem; Table 12 lists them.

10.1 Formal definitions

M0, psychometric function without memory ($d = 0$). Equations (6)–(10) with $\mathbf{w}_g = \mathbf{w}_1 = \mathbf{w}_2 = 0$: four parameters $c_1^0, \delta^0, \beta, \lambda$. Every gain is measured against this floor.

M1, one trial back ($d = 0$). M0 plus linear terms in the previous trial,

$$\hat{x}_t = s_t + \gamma_s s_{t-1} + \gamma_a^\top \text{onehot}(a_{t-1}) + \gamma_r r_{t-1} + \gamma_{ar} r_{t-1} \text{sign}(a_{t-1} - \bar{a}), \quad (15)$$

which separates “there is an effect of the previous trial” from “the effect accumulates”.

M2, criterion shift (the main cognitive model, $d = 2$). The two criteria are the state. After each trial they move towards the previous stimulus if the choice was wrong and decay towards a resting value:

$$c_{k,t} = c_{k,t-1} + \eta(1 - r_{t-1}) \text{sign}(s_{t-1} - c_{k,t-1}) \mathbb{I}[|s_{t-1} - c_{k,t-1}| < w] - \rho(c_{k,t-1} - c_k^0), \quad (16)$$

with learning rate η , window w and decay ρ , and the read-out of Eq. (9). This is the analogue of the “value” update of Ji-An et al., written for perceptual criteria.

Same problem, four operationalisations: predict $P(a_t)$ from the trial history and the current bar

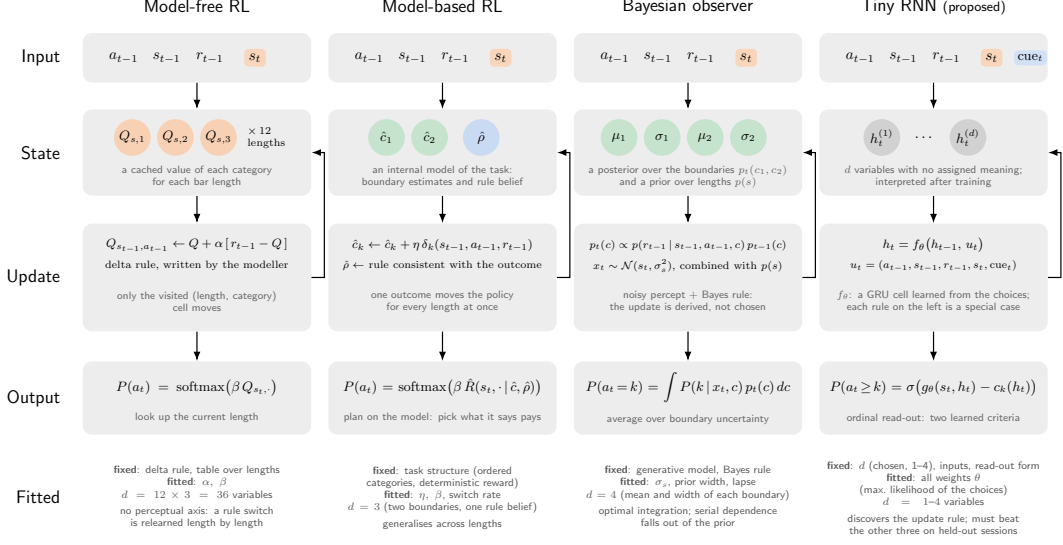


Figure 7: Four operationalisations of the same problem. All four are dynamical systems on the same inputs with the same output. They differ in what the state is, in who writes the update rule (the modeller, a derivation, or the data) and in how many variables d they carry. Following Ji-An et al., they are compared by held-out likelihood at matched d : a cognitive model that loses to a network with the same d has the wrong form of update, not too little memory.

M3, model-free reinforcement learning ($d = 36$). A table $Q_t(s, a)$ over the 12 lengths and 3 categories, updated by the delta rule $Q_t(s_{t-1}, a_{t-1}) = Q_{t-1}(s_{t-1}, a_{t-1}) + \alpha[r_{t-1} - Q_{t-1}(s_{t-1}, a_{t-1})]$ and read out by $P(a_t) = \text{softmax}(\beta Q_t(s_t, \cdot))$. It has no perceptual axis: a rule switch must be relearned length by length.

M4, Bayesian observer ($d = 4$). A Gaussian posterior over each criterion, $c_k \sim \mathcal{N}(\mu_k, \sigma_k^2)$, updated by Bayes' rule with the likelihood of the observed outcome, $p_t(c) \propto p(r_{t-1} | s_{t-1}, a_{t-1}, c) p_{t-1}(c)$; a noisy percept $x_t \sim \mathcal{N}(s_t, \sigma_s^2)$ combined with a prior $p(s)$ over lengths; and a decision that averages over the posterior, $P(a_t = k) = \int P(k | x_t, c) p_t(c) dc$. The update is derived, not chosen; serial dependence falls out of the prior.

Table 12: Models compared. All share the inputs, the output and the loss.

Model	State	d	Establishes
M0 psychometric	none	0	floor without memory
M1 one trial back	none; previous-trial terms	0	effect versus accumulation
M2 criterion shift	c_1, c_2 with Eq. (16)	2	the hand-written analogue of value
M3 model-free RL	$Q(s, a)$ table	36	reference without a perceptual axis
M4 Bayesian	posterior mean and width per criterion	4	derived update
RNN-plain	plain RNN	1–4	whether gates are needed
RNN-GRU	this specification	1–4	
Shuffled	RNN-GRU with permuted labels		floor of the NLL

10.2 Decision rule

The network “discovers something” only if it beats M1 and M2 with equal or smaller d , in test NLL, with the bootstrap interval excluding zero. If it does not beat M1, the past acts for one trial and does not accumulate. If it beats M1 but not M2, the criterion model already captures the dynamics and is the model to report.

11 Interpretation

Once trained, the network is read on the stimulus axis, not on the state axis.

1. **Criterion trajectories.** For every trial, $c_1(\mathbf{h}_t)$ and $c_2(\mathbf{h}_t)$ in degrees of visual angle, plotted against trial number with the block switches marked. This is the empirical version of the lower plots of Figure 6.
2. **Phase portrait in criterion units.** With $d = 1$: $c_k(\mathbf{h}_t)$ against $c_k(\mathbf{h}_{t-1})$, one curve per input condition (correct/error $\times$ previous category). Where a curve crosses the diagonal there is a fixed point; its slope says how much is retained. This is the plot of Ji-An et al. transferred to length units.
3. **Gates.** Distribution of z_t and r_t by condition. A mean z_t near 1 confirms short memory; r_t near 0 on the trials after a block switch confirms a reset.
4. **Input ablations.** Remove s_{t-1} , a_{t-1} , r_{t-1} or cue_t one at a time and measure the loss in NLL. This quantifies the three sources of the past: perceptual serial dependence, feedback-driven criterion shift, and rule.
5. **Perception versus decision.** Relative magnitude of $\mathbf{w}_g$ against $\mathbf{w}_1, \mathbf{w}_2$: whether the past acts on perception or on the criterion.
6. **Closed form.** If $d = 1$, fit a simple expression (piecewise linear or a low-order polynomial) to the learned update curve. If it differs from Eq. (16), it is a new cognitive rule.
7. **Chronometric check.** In the joint model, observed decision time against $|\hat{x}_t - c_k(\mathbf{h}_t)|$ for the nearest criterion, binned. The curve must fall with distance; its slope is κ in Eq. (12).

12 Predictions

Table 13: Concrete predictions and their basis. Any of these that fails is a result, not a problem with the model.

Prediction	Basis
M1 beats M0 by a small margin: the post-error effect exists	accuracy 0.69 versus 0.79
RNN $d = 1$ matches or beats M1 by a small margin; $d = 2$ does not improve on $d = 1$ in three-category sessions	weak sequential effects
Mean $z_t > 0.5$: memory of one or two trials	fixed rule, deterministic reward
Fitted c_1 near 4.6, below the nominal 5.05; c_2 near 6.4	Table 3
On the five trials after each block switch the cue-driven version beats the latent one; the latent version closes the gap within about ten trials	the cue is visible; the split moves a boundary by 0.8 deg VA, which errors reveal quickly
The between-session NLL exceeds the within-session NLL by less than the gap between M0 and the best RNN	accuracy stable at 0.73–0.81 across days
Ablating s_{t-1} costs more NLL than ablating a_{t-1}	perceptual serial dependence is the typical history effect in categorization
In the joint model $\kappa > 0$: decision times peak near the fitted criteria	chronometric effect, notebook Section 1.6
The joint model reaches the same choice NLL as the choice-only model with fewer sessions	the decision time is a continuous second observation per trial

13 Data requirements and changes to the task

The design assumes that **alternate** is the base session mode from now on. Table 14 lists what that corpus has to look like for the split of Section 9.1 to work, and what has to change in the task code.

Table 14: What the analysis needs from future sessions.

Need		Current state	What is required
Enough sessions	alternate	1 session, 6 switches	12 to 15 sessions, 50 to 70 switches in total (Section 5); ten is the floor for the latent-rule version
Variable segment length		fixed at one block (48 trials) per segment	draw alternateBlocks2cat and alternateBlocks3cat independently from $\{1, 2, 3\}$ with weights about 0.6/0.3/0.1 for every session, so the rule cannot be predicted by counting trials and switches stay frequent
Both switch directions		2→3 and 3→2 both present	keep alternating the starting rule across sessions so the two directions are balanced
A stable two-category split		marked “placeholder” in ConfigBarLengths.m	fix it: the model assumes $c_1^{2\text{cat}}$ is the same in every session
Motor bias separated from decision bias		the target layout is not exported	export the category at each of the four positions; PrevTrialDirection then becomes an input channel
Uniform schema		old sessions lack NumCategories	the notebook’s normaliser already handles it; document it in the export

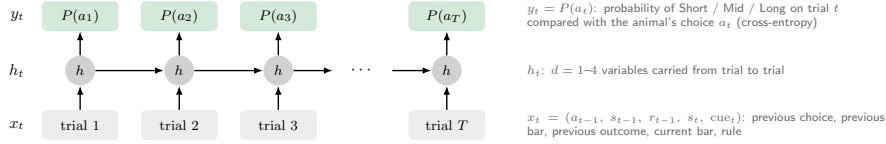
14 Relation to the within-trial networks

The companion document specifies two networks that operate inside the trial at a 10 ms step: a continuous-time rate network that reads the category out over the trial, and a GRU that produces the cursor trajectory (Figure 8). Three links tie them to the network specified here.

- The curve $s_t \mapsto P(a_t)$ of this network, evaluated at a typical $\mathbf{h}_t$, must agree with the asymptotic output $z(T)$ of the task network. If they disagree, one of the two is absorbing something that is not its own.
- The state $\mathbf{h}_t$ can serve as the initial condition of the task network on trial t , giving it history context it need not learn. This is optional and is evaluated as an ablation.
- This network is fitted first. It is the cheapest, it fixes the minimal d of history, and it says whether there is a history the other networks must respect.

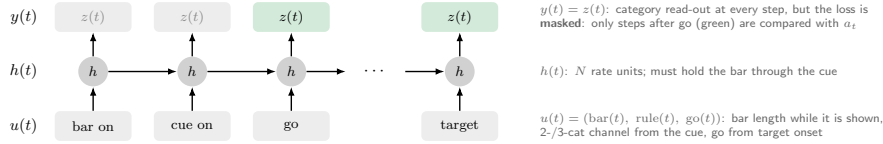
Many-to-many: every step gets an input x_t and produces an output y_t ; the loss is summed over the steps that carry a label

1. Trial-level tiny RNN (one step = one trial; the sequence is a whole session)



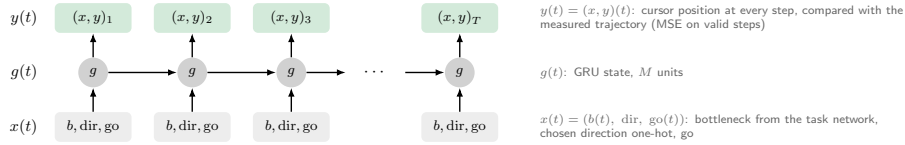
Every trial has a label, so every step contributes to the loss. Sequence length $T = 240-465$ trials.

2. Task network (one step = 10 ms; the sequence is one trial, ≈ 335 steps)



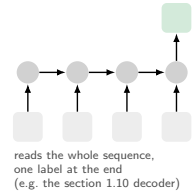
One label per trial, repeated over the post-go steps: this is where many-to-many looks like many-to-one. Keeping the per-step output is what gives the model's decision time.

3. Motor network (one step = 10 ms; same trial, aligned step by step)



A measured position at every step, so every valid step contributes. This is the purest many-to-many of the three.

Many-to-one



Many-to-many (aligned)

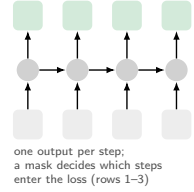


Figure 8: The three networks unrolled in time, with what x_t and y_t are in each. Row 1 is this document's network (one step per trial, every step labelled). Rows 2 and 3 are the within-trial task and motor networks (one step per 10 ms); the task network's outputs before the go signal are masked, which is where many-to-many looks like many-to-one, and the motor network has a measured position at every step. Right: the two topologies contrasted.

A Model skeleton in PyTorch

The choice loss is $-(\text{mask} * \log \text{probs}[\text{a_obs}]).\text{sum}() / \text{mask}.\text{sum}()$; the decision-time loss is the Gaussian negative log-likelihood of $\log_dt$ given μ and $\log_sd.\text{exp}()$, with the same mask, weighted by λ_{RT} . For the feedback-inferred version the two cue_t channels of $\mathbf{x}$ are set to zero; for the choice-only model the second output is ignored.

```
import torch, torch.nn as nn, torch.nn.functional as F

class TinyRNN(nn.Module):
    def __init__(self, k=9, d=1, link="logistic"):
        super().__init__()
        self.cell = nn.GRUCell(k, d)
        self.h0 = nn.Parameter(torch.zeros(d))
        self.w_g = nn.Linear(d, 1) # x_hat = s_t + w_g.h + b_g
        self.w_c = nn.Linear(d, 2) # c_1, delta (c_2 = c_1 + softplus(delta))
        self.log_beta = nn.Parameter(torch.zeros(1))
        self.lapse_raw = nn.Parameter(torch.tensor(-3.0))
        self.link = link
        # decision-time head (Sec. Decision time): mu = mu0 + kappa * D + w_dt.h
        self.w_dt = nn.Linear(d, 1)
        self.kappa = nn.Parameter(torch.zeros(1))
        self.log_sd = nn.Parameter(torch.zeros(1))

    def cdf(self, u):
        if self.link == "logistic":
            return torch.sigmoid(u)
        return 0.5 * (1 + torch.erf(u / 2 ** 0.5)) # cumulative Gaussian

    def forward(self, x, s): # x: (T, B, k), s: (T, B)
        T, B, _ = x.shape
        h = self.h0.expand(B, -1)
        out, mu = [], []
        for t in range(T):
            h = self.cell(x[t], h)
            xhat = s[t] + self.w_g(h).squeeze(-1)
            c = self.w_c(h)
            c1 = c[:, 0]
            c2 = c1 + F.softplus(c[:, 1])
            beta = self.log_beta.exp()
            lam = 0.1 * torch.sigmoid(self.lapse_raw)
            p1 = (1 - lam) * self.cdf(beta * (xhat - c1)) + lam / 2 # P(a >= 1)
            p2 = (1 - lam) * self.cdf(beta * (xhat - c2)) + lam / 2 # P(a >= 2)
            probs = torch.stack([1 - p1, p1 - p2, p2], dim=-1).clamp_min(1e-6)
            out.append(probs)
            D = -torch.minimum((xhat - c1).abs(), (xhat - c2).abs()) # Eq. (dt-D)
            mu.append(self.w_dt(h).squeeze(-1) + self.kappa * D)
        return torch.stack(out), torch.stack(mu) # (T, B, 3), (T, B)
```